Task Management System

A modern, full-featured task management application built with React, featuring role-based authentication, real-time status updates, and interactive data visualization.



Features

Admin Dashboard

- **Task Management**: Create, edit, delete tasks with validation
- **Analytics Dashboard**: Real-time statistics with pie & bar charts
- **Advanced Filtering**: Search by title, filter by status/employee
- **Performance Tracking**: Employee-wise completion rates
- **Responsive Design**: Optimized for all screen sizes

Employee Dashboard

- **Task Overview**: View assigned tasks with status cards
- **Status Management**: Update task status (New → Active → Completed/Failed)
- **Visual Statistics**: Task count by status with color-coded cards
- **Task Details**: View task descriptions, due dates, and assigned by info

Authentication System

- **Role-Based Access**: Admin & Employee roles with protected routes
- **Persistent Sessions**: Local storage-based authentication
- **Route Protection**: Unauthorized access prevention

Tech Stack

- **Frontend Framework**: React 18.3.1
- **Styling**: Tailwind CSS 3.4.17
- **Charts**: Recharts 2.15.0
- **Routing**: React Router DOM 7.1.1
- **Build Tool**: Vite 6.0.5
- **State Management**: Context API

Installation

Prerequisites

- Node.js (v16 or higher)
- npm or yarn

Steps

1. **Clone the repository**

```
git clone https://github.com/YOUR_USERNAME/task-management-system.git
cd task-management-system
```